# ZK-Proof-Based Reputation Claims in a Decentralized Lending System

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# Project Approval & Guide Signature

**"ZK-Proof-Based Reputation Claims in a Decentralized Lending System"**

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# Introduction

- **Evolution of Blockchain Consensus**
  Blockchain systems have evolved from energy-intensive PoW to efficient models like PoS and PoR.
- **Proof-of-Reputation (PoR)**
  PoR selects validators based on behavioral history and trust rather than computation or stake.
- **Advantages of PoR**
  Provides energy efficiency, trust-driven validation, and improved security in decentralized systems.
- **Cold-Start Limitation**
  New nodes lack reputation, preventing participation in consensus and network growth.
- **Our Contribution**
  We propose ColdStart-PoR, a structured protocol enabling secure and fair entry of new nodes.

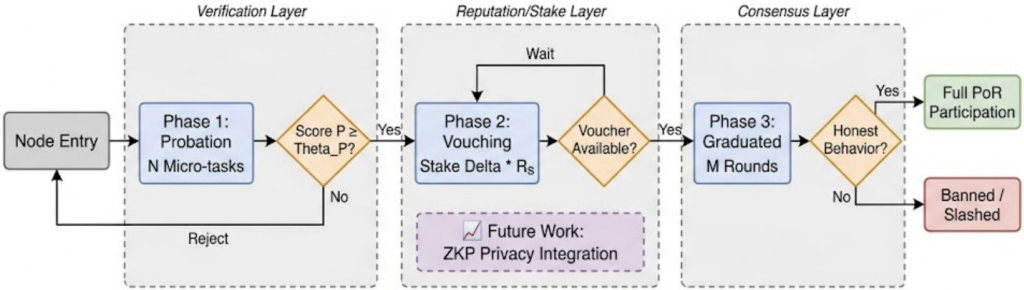
# Literature Survey

- **Evolution of Consensus Mechanisms** Blockchain evolved from PoW/PoS to Proof-of-Reputation (PoR), where validators are selected based on behavioral trust.
- **Limitations of Existing PoR Systems** Assume pre-existing reputation or genesis assignment $\rightarrow$ leads to centralization and Sybil attacks.
- **Blockchain-based Credit & Lending Systems** Enable trustless finance but lack strong reputation bootstrapping mechanisms.
- **Zero-Knowledge Proofs for Privacy** ZKPs (zk-SNARKs / zk-STARKs) allow verification without revealing sensitive data.
- **Privacy Limitations in Existing Systems** Current KYC systems expose user data or lack secure compliance encoding.
- **Layer-2 Scalability Solutions** zk-Rollups improve scalability and cost but are not

# Problem Statement

- **Cold-Start Issue in PoR**
  New nodes enter the network with zero reputation and cannot participate in consensus.
- **Circular Dependency**
  Participation is required to gain reputation, but reputation is required to participate.
- **Limitations of Existing Methods**
  Genesis assignment introduces centralization, while uniform reputation enables Sybil attacks.
- **Security Risks**
  Weak entry mechanisms allow adversaries to create multiple fake identities.
- **Core Objective**
  Develop a decentralized, secure, and fair mechanism for reputation bootstrapping.

# System Architecture



# Backend - Sybil Cost Analysis

- **Objective**
  Evaluate resistance against Sybil attacks using cost modeling.
- **Parameters**
  $\delta = 15\%$, $\tau_v = 40\% \rightarrow$ Cost per Sybil $= \delta \cdot \tau_v = 600$ BPS
- **Observation**
  Total attack cost increases linearly with number of identities ($k$).
- **Comparison**
  Existing approaches (Genesis / Uniform) allow near-zero cost attacks.
- **Conclusion**
  ColdStart-PoR enforces **linear cost growth**, making large-scale Sybil attacks infeasible.

# Backend - Protocol Execution Summary

- **Simulation Setup**
  Network with 4 nodes executed over multiple rounds.
- **Key Parameters**
  $\delta = 15\%$, $\alpha = 50\%$, $\theta_P = 90\%$, $\tau_v = 40\%$, $\lambda = 80\%$
- **Process**
  Nodes undergo probation, vouching, and graduated participation phases.
- **Result**
  Nodes successfully transition to full participation with stable reputation.
- **Insight**
  Protocol maintains fairness and security with minimal overhead.

# Backend - Output Visualization

```
Verifying Sybil cost model (Proposition 1):
δ = 1500 BPS, τ_v = 4000 BPS
Min cost per Sybil = δ·τ_v = 600 BPS

k  | Total cost (BPS) | vs. Genesis/Uniform (O(1)=0)
---|-----------------|----------------------------
1  | 600              | O(k·600) = linear ✓
5  | 3000             | O(k·600) = linear ✓
10 | 6000             | O(k·600) = linear ✓
25 | 15000            | O(k·600) = linear ✓
50 | 30000            | O(k·600) = linear ✓
100| 60000            | O(k·600) = linear ✓
```

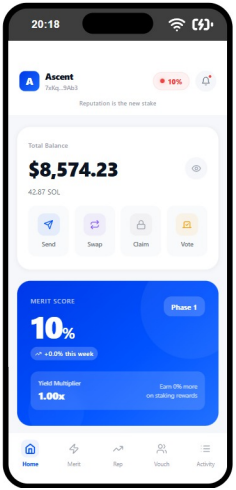
Sybil Cost Analysis Output

```
═══════════════════════════════════════════════════
ColdStart-PoR Protocol Summary
═══════════════════════════════════════════════════
Network round:  3
Total nodes:    4
Parameters:     δ=15% α=50% θP=90% τv=40% λ=80%

Genesis node:   R = 7000 BPS | Phase: Full
Graduated node: R = 5148 BPS | Phase: Full
═══════════════════════════════════════════════════
  ✓ 9. Protocol summary — fetch and display final state
```

Protocol Execution Summary Output

# Frontend Visualization




User interface showing reputation score, balance, and participation features

# Frontend Overview

- **User Interface**
  Provides an intuitive dashboard for interacting with the blockchain system.
- **Core Features**
  Displays user balance, reputation score, and participation phase in real-time.
- **User Actions**
  Enables operations such as voting, staking, and tracking merit/reputation.
- **Web3 Integration**
  Connects to blockchain via wallet and transaction APIs (e.g., JSON-RPC).
- **Visualization**
  Clearly represents trust (reputation) as a measurable and dynamic metric.

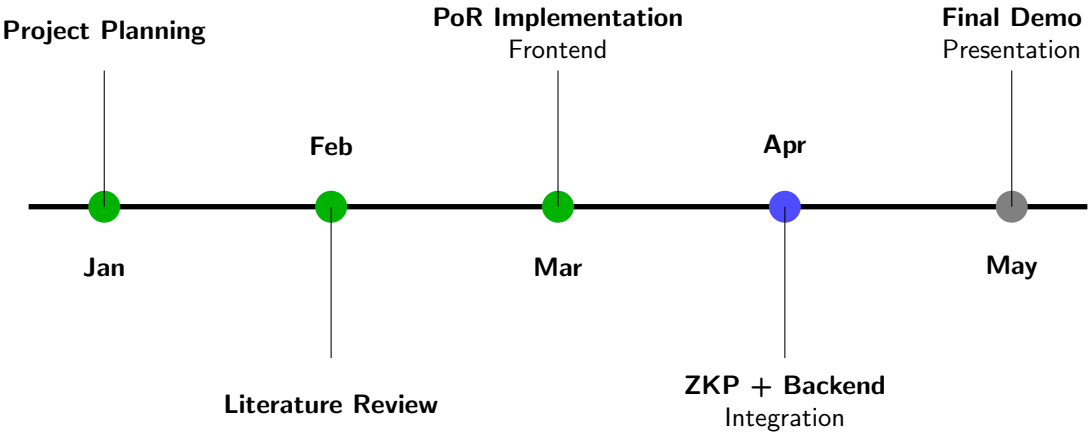
# System Extension - ZK Proofing

- **Objective**
  Enable privacy-preserving reputation using Zero-Knowledge Proofs (ZKPs).
- **ZK Mechanism**
  Use zk-SNARK/zk-STARK to prove reputation validity without revealing raw data.
- **Hardhat Implementation**
  Develop and test ZK-enabled smart contracts using Hardhat framework.
- **Proof Verification**
  Integrate on-chain verification of proofs for trustless validation.
- **Security Benefit**
  Prevents exposure of user identity and vouching relationships.

# System Extension - ZK Proofing

- **ZK Wallet Integration**
  Extend frontend to support proof generation via user wallet.
- **Seamless User Flow**
  Allow users to submit ZK proofs during participation without extra complexity.
- **API Integration**
  Connect frontend with ZK-enabled smart contracts via Web3/JSON-RPC.
- **UI Enhancements**
  Display verified reputation without exposing sensitive data.
- **End-to-End System**
  Achieve a fully decentralized, privacy-preserving reputation ecosystem.

# Project Timeline



# Contributions

| Member | Contribution |
|---|---|
| Abhijith S | ZK-proof module design, Hardhat Integration |
| Amal Anand | Frontend UI, Web3 integration |
| Heman Sakkthivel | Backend PoR logic, reputation engine |
| Roshan R | Smart contract interaction, testing |
| Vaishnav Raj M | System integration, API coordination |

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